FACE Prep

Test Run

Answer the Following questions

1. $92 \times 69 =$ _____

2. $\sqrt[3]{12167} =$ _____

3. $1476 \times 452 =$

   a. 667142 b. 667162

   c. 667152 d. 667182

4. $\sqrt{23104} =$ _____

5. $372 \times 9 =$ _____

6. (20% of 40) × 111 × 125 = _____

7. $91 \times 109 \times 11 =$

8. $122 + 182 + 2 \times 12 \times 18 =$ _____

9. $\sqrt{1+ (20 \times 21 \times 22 \times 23)} =$ _____

   a. 441 b. 451

   c. 461 d. 471

10. Find the sum of first 231 odd numbers.

11. $\sqrt{75^2 + 100^2} =$ _____

12. Arrange the given fractions in ascending order:

$\frac{112}{114}, \frac{118}{120}, \frac{114}{114}, \frac{122}{124}, \frac{116}{118}$

13. $\sqrt[3]{474552} =$ _____

14. $78 \times 82 =$ _____

15. $\frac{164 \times 64.5 + 22.1}{2.5} =$ _____

    a. 2425.3 b. 2429.3

    c. 2426.3 d. 2325.3

16. $63 \times 231 =$ _____

17. Arrange the following fractions in descending order:

$\frac{57}{1185}, \frac{71}{1210}, \frac{109}{910}$

18. $3003 \times 3097 =$ _____

19. $72^3 =$

    a. 373678 b. 367248

    c. 373248 d. 353428

20. $744 \times 839 =$ _____

21. $842 \times 273 =$ _____

22. $\sqrt{62 \times 4216 \times 68} =$ _____

23. $3 \times 7 \times 11 \times 13 \times 17 =$ _____

a. 51501 b. 51051

c. 51061 d. 51151

24. $347 \times 351 =$ _____

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FACE Prep

Concept 1

1.1 SAW Method

1. 69 × 7 = _____
   =(__-__) × 7
   =___ × 7 - ___×7
   =___-___

2. 72 × 11= _____
   =(__+__) × 72
   =___ + ___
   =_____

3. 12 × 23 = _____

4. 43 × 12 = _____

5. 31 × 11 = _____

6. 97 × 17 = _____

7. 23 × 19 = _____

8. 101× 12 = _____

9. 47 × 12 = _____

10. 43 × 51 =_____

11. 36 × 21 = _____

12. 67 × 22 = _____

13. 17 × 31 = _____

14. 88 × 99 = ____

15. 14 × 29 = _____

16. 73 × 41 = _____

1.2 Let's do a "Total Recall"

1. 11^2 = _____

2. 12^2 = _____

3. 13^2 = _____

4. 14^2 = _____

5. 15^2 = _____

6. 16^2 = _____

7. 17^2 = _____

8. 18^2 = _____

9. 19^2 = _____

10. 21^2 = _____

11. 22^2 = _____

12. 23^2 = _____

13. 24^2 = _____

14. 25^2 = _____

15. 7^3 = _____

16. 8^3 = _____

17. 9^3 = _____

18. 10^3 = _____

19. 11^3 = _____

20. 12^3 = _____

1.3 0z Method

1. 124 × 5 = _____
   124 ×—= _____ × 10
   = _____

2. 36 × 25 = _____
   36 × — = _____ × 100
   = _____

3. 24 × 125 = _____

4. 48 × 75 = _____

5. 4844 × 25 = _____

6. 786 × 50 = _____

7. 88 × 175 = _____

8. $432 \times 125 =$ _____

9. $123 \times 11 =$ _____

10. $324 \times 11 =$ _____

11. $1211 \times 22 =$ _____

12. $244 \times 55 =$ _____

13. $8040 \times 275 =$ _____

14. $640 \times 375 =$ _____

15. $22 \times 6 \times 50.5 =$ _____

16. $75 \times 58 \times 44 =$ _____

Concept 2

2.1 Mirrors method

Numbers between 40 -60

1. $47^2 =$ _____

2. $41^2 =$ _____

3. $56^2 =$ _____

4. $49^2 =$ _____

5. $52^2 =$ _____

6. $43^2 =$ _____

7. $59^2 =$ _____

8. $54^2 =$ _____

Numbers between 90 -110

1. $97^2 =$ _____

2. $106^2 =$ _____

3. $99^2 =$ _____

4. $102^2 =$ _____

5. $95^2 =$ _____

6. $93^2 =$ _____

7. $109^2 =$ _____

8. $108^2 =$ _____

2.2 Hi-Fi method

Squares of Numbers ending in 5

1. $25^2 =$ _____
Last 2 digits of the square =
Remaining digits = __×__
Final Answer = _____

2. $75^2 =$ _____
Last 2 digits of the square =
Remaining digits = __×__
Final Answer = _____

3. $105^2 =$ _____

4. $795^2 =$ _____

5. $405^2 =$ _____

6. Last 2 digits of $155 \times 185 =$ _____

7. Last 2 digits of $785 \times 805 =$ _____

8. Last 2 digits of $1695 \times 1415 =$ _____

Finding the last 2 digits of Squares of Numbers

9. $91^2 =$ _____
Nearest multiple of 100 = _____
Deviation = _____
Last 2 digits of the deviation^2 = _____

10. $127^2 =$ _____
Nearest multiple of 100 = _____
Deviation = _____
Last 2 digits of the deviation^2 = _____

11. $79^2 =$ _____

12. $724^2 =$ _____

13. $996^2 =$ _____

14. $73437^2 =$ _____

15. 2617^2 = _____

16. 13235^2 = _____

2.3 Minions method

Squares of any numbers between 1 – 1000

1. 67^2 = _____

2. 36^2 = _____

3. 72^2 = _____

4. 83^2 = _____

5. 64^2 = _____

6. 123^2 = _____

7. 221^2 = _____

8. 472^2 = _____

9. 558^2 = _____

19. 982^2 = _____

Concept 3

3.1 Square Roots of perfect square numbers

1. $\sqrt{1296}$ = _____
Last Digit = _____
Last digit of square root __ or __
The first 2 digits of the given number is 12 and lie between $_^2$ and $_^2$
By High5 method, unit digit is __
Answer = __

2. $\sqrt{1369}$ = _____
Last Digit = _____
Last digit of square root __ or __
The first 2 digits of the given number is 13 and lie between $_^2$ and $_^2$
By High5 method, unit digit is __
Answer = __

3. $\sqrt{3025}$ = _____

4. $\sqrt{2116}$ = _____

5. $\sqrt{5184}$ = _____

6. $\sqrt{5776}$ = _____

7. $\sqrt{6561}$ = _____

8. $\sqrt{7921}$ = _____

9. $\sqrt{9409}$ = _____

10. $\sqrt{7921}$ = _____

11. $\sqrt{12100}$ = _____

12. $\sqrt{12996}$ = _____

13. $\sqrt{18225}$ = _____

14. $\sqrt{26244}$ = _____

15. $\sqrt{34225}$ = _____

16. $\sqrt{207936}$ = _____

17. $\sqrt{301401}$ = _____

3.2 Cube Roots of perfect cube numbers

1. $\sqrt[3]{1728}$ = _____
Last Digit = _____
Last digit of cube root = ___
The first 2 digits lie between $_^3$ and $_^3$
Answer = __

2. $\sqrt[3]{5832}$ = _____
Last Digit = _____
Last digit of cube root = ___
The first 2 digits lie between $_^3$ and $_^3$
Answer = __

3. $\sqrt[3]{17576}$ = _____

4. $\sqrt[3]{42875}$ = _____

5. $\sqrt[3]{91125}$ = _____

6. $\sqrt[3]{110592}$ = _____

7. $\sqrt[3]{132651}$ = _____

8. $\sqrt[3]{205379}$ = _____

9. $\sqrt[3]{300763}$ = _____

10. $\sqrt[3]{512000}$ = _____

11. $\sqrt[3]{592704}$ = _____

12. $\sqrt[3]{912673}$ = _____

13. $\sqrt[3]{1404928}$ = _____

14. $\sqrt[3]{2571352}$ = _____

15. $\sqrt[3]{4492125}$ = _____

16. $\sqrt[3]{11239424}$ = _____

Concept 4

4.1 The Golden eye method

1. 42 × 48 = _____
Last 2 digits of the product =
First 2 digits of the product =
Answer = _____

2. 23 × 27 =
Last 2 digits of the product =
First 2 digits of the product =
Answer = _____

3. 16 × 14 = _____

4. 54 × 56 = _____

5. 42 × 62 = _____

6. 27 × 87 = _____

7. 16 × 96 = _____

8. 66 × 46 = _____

9. 890 × 810 = _____

10. 211 × 289 = _____

11. 7030 × 7070 = _____

12. 8699 × 8601 = _____

13. 2010 × 2090 = _____

14. 4589 × 4511 = _____

15. 10992 × 10008 = _____

16. 53002 × 53998 = _____

4.2 Base Method Multiplication

1. 47 × 52 = _____

2. 94 × 103 = _____

3. 98 × 102 = _____

4. 96 × 96 = _____

5. 208 × 198 = _____

6. 397 × 402 = _____

7. 503 × 507 = _____

8. 504 × 498 = _____

9. 607 × 594 = _____

10. 891 × 902 = _____

11. 554 × 548 = _____

12. 999 × 999 = _____

4.3 Multiplication from squares

1. 48 × 52 = _____
Difference between the numbers = _____
Is the difference (d) between them even?
'X' is $48 + \frac{d}{2}$ or $52 - \frac{d}{2}$ = _____
$X^2 - (\frac{d}{2})^2$ =

2. 84 × 76 = _____
Difference between the numbers = _____
Is the difference (d) between them even?
'X' is $76 + \frac{d}{2}$ or $84 - \frac{d}{2}$ = _____
$X^2 - (\frac{d}{2})^2$ =

3. 104 × 96 = _____

4. 78 × 92 = _____

5. 33 × 57 = _____

6. $87 \times 113 =$ ____

7. $69 \times 43 =$ ____

8. $65 \times 85 =$ ____

9. $127 \times 95 =$ ____

10. $321 \times 345 =$ ____

4.4 Crisscross method

1. $62 \times 87 =$ ____

2. $36 \times 74 =$ ____

3. $28 \times 57 =$ ____

4. $888 \times 96 =$ ____

5. $343 \times 643 =$ ____

6. $489 \times 738 =$ ____

7. $123 \times 321 =$ ____

8. $6321 \times 9011 =$ ____

9. $9999 \times 6666 =$ ____

10. $1234 \times 4321 =$ ____

Concept 5

5.1 Comparing Fractions

Arrange the following fractions in ascending order

1. $\frac{1}{5}, \frac{3}{5}, \frac{2}{5}, \frac{8}{5}, \frac{6}{5}$
2. $\frac{5}{3}, \frac{5}{7}, \frac{5}{2}, \frac{5}{9}, 5$
3. $\frac{1}{2}, \frac{4}{5}, \frac{3}{4}, \frac{8}{9}, \frac{6}{7}$
4. $\frac{11}{7}, \frac{15}{11}, \frac{7}{3}, \frac{6}{2}$
5. $\frac{4}{7}, \frac{15}{21}, \frac{12}{14}, \frac{9}{21}, \frac{2}{14}$
6. $\frac{1}{5}, \frac{3}{5}, \frac{2}{5}, \frac{8}{5}, \frac{6}{5}$
7. $\frac{10}{8}, \frac{20}{6}, \frac{30}{12}, \frac{100}{70}, \frac{55}{11}$
8. $\frac{9}{11}, \frac{11}{13}, \frac{17}{19}, \frac{16}{18}, \frac{15}{17}, \frac{12}{14}$
9. $\frac{9}{11}, \frac{7}{9}, \frac{4}{13}, \frac{5}{17}$
10. $\frac{17}{5}, \frac{19}{8}, \frac{23}{13}, \frac{18}{5}$

5.2 Butterfly Method

1. $\frac{16}{19}, \frac{17}{22}$
A = 16 × 22
B = 17 × 19
If A>B, I fraction is greater
If A<B, II fraction is greater

2. $\frac{11}{15}, \frac{21}{26}, \frac{14}{17}$
Compare the first two fractions
A = 14 × 15 = ____
B = 17 × 11 = ____
Compare the II and III fractions
C = 11 × 26 = ____
D = 21 × 15 = ____

3. $\frac{52}{27}, \frac{33}{17}$

4. $\frac{11}{13}, \frac{14}{17}, \frac{9}{11}$

5.3 Inception Method

1. Which of the following is greater?
$\frac{140}{287}, \frac{291}{530}$
The percentage increase in the Nr =
The percentage increase in the Dr =
If the % increase in the Nr > % increase in the Dr, fraction II is greater
Else, fraction I is greater

2. Which of the following is greater?
$\frac{142}{380}, \frac{432}{795}$
The percentage increase in the Nr =
The percentage increase in the Dr =
If the % increase in the Nr > % increase in the Dr, fraction II is greater
Else, fraction I is greater

3. Which is the greater of the fractions $\frac{1230}{920}, \frac{1074}{850}$?

4. Which of the following is greater?
$\frac{389}{150}, \frac{1234}{737}$

5.4 Percentages

1. 50% of 40 = _____
2. 22% of 90 = _____
3. 70% of 20 = _____
4. 45% of 60 = _____
5. 82% of 150 = _____
6. 11.11% of 81 = _____
7. 16.66% of 42 = _____
8. 9.09% of 99 = _____
9. 28.57% of 98 = _____
10. 31.25% of 64 = _____
11. 69.75% of 256 = _____
12. 72% of 44.44 = _____
13. 57.142% of 81 = _____
14. 66.66% of 333 = _____

5.5 LCM and HCF

Find the HCF of the following numbers

1. 34, 85, 170
Prime factors of 34 =
Prime factors of 85 =
Prime factors of 170 =
HCF = _____

2. 27, 36, 63, 72
Prime factors of 27 =
Prime factors of 36 =
Prime factors of 63 =
Prime factors of 72 =
HCF = _____

3. 2365, 1315

4. 432, 31, 585, 692, 969

5. 123123, 131131

6. 646, 1292

Find the LCM of the following numbers

1. 5, 15, 12

2. 10, 12, 15

3. 42, 56

4. 64, 68

5. 30, 75, 50

6. 56, 44, 53

Concept 6

6.1 Cast Away Method

Exercise 1
Find the remainder when the following numbers are divided by 9, 8 and 11

	51480	453228120	5909090994	6245736475	318438351
/9					
/8					
/11					

Exercise 2
Questions solvable by divisibility rule for 9

1. 978 × 628
   a. 614183 b. 614184
   c. 624284 d. 614284

2. 9696 × 3636
   a. 35254656 b. 35154656
   c. 35254655 d. 35254646

3. Find the product of 9 × 99 × 999 × 9999
   a. 8900199881 b. 8900199891
   c. 8900199882 d. 8900199811

4. 134568 × 90
   a. 12111220 b. 12311120
   c. 12111120 d. 12211120

Exercise 3
Questions solvable by divisibility rule for 8

1. 98 × 72
   a. 7054 b. 7057
   c. 7056 d. 7888

2. 131072 × 511
   a. 66977782 b. 66977791
   c. 66977892 d. 66977792

3. 50904 × 44256
   a. 2252807514 b. 2252807424
   c. 2252807344 d. 2252807264

Exercise 4
Questions solvable by divisibility rule for 11

1. 152 × 228
   a. 36023 b. 26024
   c. 36024 d. 36124

2. 783 × 552
   a. 432226 b. 441216
   c. 432296 d. 432216

3. 59994 × 53328
   a. 3199360042 b. 3199359952
   c. 3118360032 d. 3199360032

4. 1458 × 6974
   a. 10168092 b. 10168292
   c. 11168092 d. 10169096

Exercise 5 – Fractions

1. Find the value of $(\frac{2}{13}) \times (\frac{3}{34}) \times 2652$
   a. 36 b. 26
   c. 16 d. 46

2. $(\frac{4}{7}) \times (\frac{2}{7}) \times (\frac{3}{7}) \times 5831$
   a. 409 b. 408
   c. 508 d. 509

3. $\frac{57}{4} \times \frac{75}{7} \times \frac{21}{10}$
   a. 321.625 b. 310.625
   c. 320.625 d. 323

Exercise 6 – Decimals

1. 12.123 × 34.345
   a. 416.364435 b. 415.354535
   c. 420 d. 415.364435

2. 9.9 × 1.1 × 2.2
   a. 23.948 b. 21.958
   c. 22.958 d. 23.958

Assorted applications

1. 2/3rd of the balls in a bag are blue, the rest are pink. If 5/9th of the blue balls and 7/8th of the pink balls are defective, find the total number of balls in the bag, given that the number of non-defective balls is 146.
   a. 276 b. 649
   c. 432 d. 578

2. In a city there are few engineering, MBA and CA candidates. Sum of 4 times of the engineering, three times of the MBA and 5 times of the CA candidates is 3650. Also sum of three times MBA candidates is equal to two times CA candidates. In total, how many MBA candidates are there in the city?
   a. 200 b. 300
   c. 450 d. 400

3. Arrange the people in the increasing order of their speeds.

Person	A	B	C	D
Distance (in kms)	213	50	217	51
Time (in hours)	217	51	221	52

4. Solve the expression.

$$\frac{77! \times (77! - 2 \times 54!)^3}{(77! + 54!)^3} + \frac{54! \times (2 \times 77! - 54!)^3}{(77! + 54!)^3}$$

   a. 2(77! + 54!) b. (77! - 54!)
   c. (77! + 54!) d. 2(77! - 54!)

5. Evaluate the following expression.

$$\frac{(44444445 \times 88888885 \times 44444442 \times 44444438)}{44444444^2}$$

   a. 88888883 b. 88888883
   c. 88888888 d. 44444443

Note: All the numbers in the question and options are 8 digit numbers

6. There are 14 digits of credit card numbers to be filled. Each of the below 3 boxes contains continuous digits of 18 as sum. Given: 4th digit is 7 and 7th digit is X. What is the value of X?

a. 1 b. 7
c. 4 d. 2

7. Product of 2 numbers is 7221 and their sum is 170. Find the numbers.
Hint: Use your golden eye.

8. Areas of 2 squares are 15376 and 15876 respectively. A rectangle is constructed with the sides of the 2 squares as its dimensions. Find the area of the rectangle so formed.

9. 1a3b23 is divisible by 182 and 121. Find a and b.

10. Find the greatest 4 digit number which is a
a. A perfect square
b. A perfect cube

11. Stove Bobs is crazy about 'ORANGE' products. Recently, an ORANGE Ibat2, costing Rs. 54880 was launched. He, being an ORANGE maniac, isn't able to resist from buying it. He has been saving money for 3 years in a bank which pays an interest 40% p.a. compounded annually. What is the minimum amount Stove should have started with so that he gets his wish fulfilled?
a. Rs. 18,000 b. Rs. 19,000
c. Rs. 20,000 d. Rs. 21,000

12. Your college principal has invested a principal of Rs. 5000 in OCOCI bank. The bank is interested to pay an interest of 18% per annum compounded annually. If the principal invested in the bank for 2 years, what will happen to the principal at the end of 2 years?
a. Rs. 6562 b. Rs. 6762
c. Rs. 6862 d. Rs. 6962